



Opto-electronic simulation of silicon waveguide integrated single photon avalanche diodes

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Abstract: Single-photon avalanche diodes (SPADs) integrated into photonic integrated circuits (PICs) can offer a promising alternative to cryogenic superconducting detectors, combining CMOS compatibility with room-temperature operation. A comprehensive optoelectronic multiphysics simulation framework to model waveguide integrated SPADs is shown here. Based on imec's isipp50G technology, simulated devices show the potential of peak photon detection efficiencies (PDE) above 75% at 600 nm, when coupled with a SiN signal waveguide. This work establishes a scalable methodology for design technology co-optimization (DTCO) of SPADs for integrated photonics and identifies critical pathways towards performance improvements, mainly in dark count rate (DCR) reduction, paving the way for their full optimization and integration into quantum photonic circuits.

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1. Introduction

Recent developments of integrated photonic platforms have led to the demonstration of quantum operations on chip, enabled by the advent of mature low-loss photonics platforms based on silicon on insulator (SOI), silicon nitride (SiN) and III-V materials such as GaAs, as well as their potential combination through hybrid integration [1–3]. On-chip detection of single photons has been made possible by resorting to the much acclaimed superconducting nanowire single-photon detectors (SNSPDs) [4–6]. Yet, their reliance on cryogenic operation (< 4 K) introduces significant barriers to scalability, due to prohibitive energy costs and, more importantly, challenging compatibility with large scale photonic integrated circuits (PICs) [7]. This key limitation underscores the need for alternatives that are compatible with monolithic integration and operate at less extreme temperatures.

Single-photon avalanche diodes (SPADs) present a compelling solution. These semiconductor based devices operate in the Geiger mode, i.e., biased above the breakdown voltage and have been widely adopted in free space quantum photonics, LiDAR and other applications such as biomedical imaging due to their room (or Peltier cooled) temperature operation, cost efficiency and CMOS compatible fabrication [8,9]. However, little research has been made on bringing these devices to waveguide integrated solutions. Efforts on this have been largely confined to infrared wavelengths (Ge-on-Si) such as in [10,11], leaving a gap in the visible spectrum. This is a critical gap, essential for quantum technologies such as color centers or quantum dots. While silicon's opacity in the visible precludes its use in passive components at such wavelengths, its strong absorption characteristics make it an ideal candidate for detectors when heterogeneously integrated with an appropriate supporting platform. A critical advantage versus traditional SPAD (non waveguide) implementations lies in the orthogonal alignment of optical propagation and biasing field vectors (Fig. 1), which decouples the traditional trade-off between photon detection

efficiency (PDE) and timing jitter in free-space SPADs. In conventional architectures, thicker active regions improve efficiency but enlarge diffusion-dominated areas, degrading jitter [12]. Waveguide integration circumvents this by confining light orthogonally to the depletion region, enabling a good overlap of the multiplication and absorption regions.

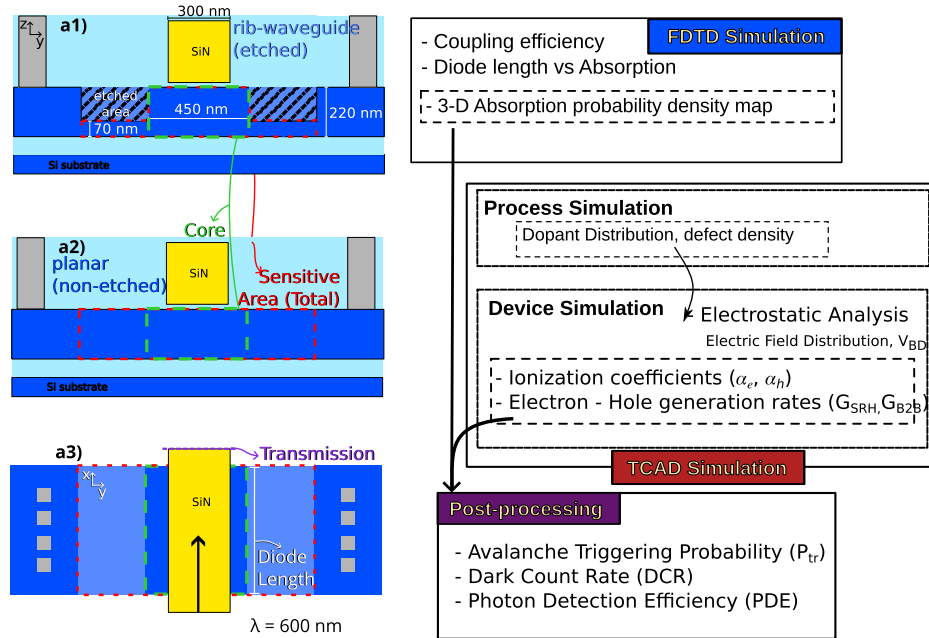


Fig. 1. a) A schematic view of the optical geometries tested, including the silicon diode (dark blue) separated from the SiN waveguide (yellow by a 50 nm layer of silica (light blue)), and electrical vias (grey). The dotted red line delimits the sensitive area, and the green dotted line delimits the core of the junction; a1) cross-section for a rib-waveguide diode with etched sides; a2) cross-section for a planar diode; a3) top-view b) Schematic representation of the adopted simulation strategy.

Recent efforts to develop visible wavelength waveguide Si SPADs have taken approaches that differ in coupling scheme and overall technology. In [13,14], SOI rib waveguides are proposed, interfaced to a supporting SiN waveguide through butt-coupling. These show promising results, although face fabrication complexities. In [15], SiN waveguides couple evanescently with a mesa Si SPAD. Although performance is limited by the mesa approach, leading to optical losses to the substrate. The work detailed in [16] describes a monolithic approach, taking advantage of a larger device volume and well established process node to reveal high PDE and low DCR. In this work, we propose to build on these advances and put forth a design of evanescently coupled rib waveguide SPADs, leveraging the recent development of hybrid photonic platforms made of both silicon and silicon photonics [17]. Low loss SiN is still the supporting platform of choice, where hybrid integration with Si based photodetectors has been previously successfully demonstrated [18]. A comparison between the optical performance of planar and etched approaches is done, and a schematic of the structure is presented in Fig. 1(a).

The proposed device is modeled while addressing key limitations in prior simulation frameworks. Earlier studies simplified optical propagation to fundamental mode analysis or relied on computationally intensive Monte Carlo methods for electrical modeling [13]. Our approach combines three-dimensional finite-difference time-domain (FDTD) optical simulations with technology computer-aided design (TCAD) electrical modeling and Python-based post-processing,

enabling a holistic optoelectronic analysis of absorption profiles, breakdown dynamics, and the relevant performance metrics : PDE and dark count rate (DCR). As an FDTD solver, Lumerical Solutions FDTD was employed, while the chosen TCAD softwares for device and process simulation was Synopsys' Sentaurus Device and Process packages. This results in a more tractable simulation process, as shown in the flowchart presented in Fig. 1(b). Indeed, this simulation methodology is a natural progression of recent modeling efforts, being adopted by others as a fast and efficient tool to assess the performance of SPAD devices with complex optical dynamics [19]. The doping levels are not optimized for SPAD design, as this falls outside the scope of our study. Instead, by leveraging existing waveguide PN modulator designs, we enable experimental validation of the presented methodology through DCR measurements of these devices.

2. Waveguide SPAD design

2.1. Geometry and optical performance

The efficient coupling between devices is an essential aspect of integrated photonic design. In this case, the method of coupling light to the SPAD can determine the performance of the device. The proposed scheme relies on the evanescent coupling from the signal waveguide (SiN, represented in yellow in Fig. 1(a)) to the waveguide SPAD.

The SPAD is based on a silicon-on-insulator (SOI) platform (isipp50G) comprising a 220 nm thick silicon (Si) layer sandwiched between top and bottom silicon dioxide (SiO_2) cladding layers. A rib-waveguide geometry with a socket-etched design, where the rib regions are etched to a thickness of 70 nm is compared to a non-etched design. The waveguide core has a width of 450 nm. The silicon nitride (SiN) input waveguide is envisioned to maintain fixed cross-sectional dimensions of $300 \text{ nm} \times 300 \text{ nm}$, embedded in SiO_2 , separated from the Si structure by a 50 nm oxide gap.

Due to the geometric disparity between the uniform SiN waveguide and the etched Si-SPAD, a modal mismatch occurs at their interface, disrupting optical mode propagation. This mismatch presents an opportunity for future optimization—for instance, through adiabatic tapering of the SiN waveguide—to improve coupling efficiency and enhance photon absorption within the Si-SPAD, thereby enabling shorter active device lengths. Nonetheless, as a starting point, the large index contrast between Si and SiN should result in a fairly efficient coupling even without a waveguide tapering.

The coupling scheme must achieve two key objectives: transfer light from the SiN waveguide to the SPAD and confine the absorption to a small volume to maximize overlap between the optical field and the high-electric-field region (space charge region, SCR) of the SPAD. The latter is critical for SPAD performance as avalanche triggering probability is significantly higher when absorption occurs directly within the SCR, minimizing carrier diffusion delays and therefore minimizing jitter. While the optical coupling efficiency depends on waveguide design, the SCR overlap is also influenced by the junction architecture.

For our modeling, we assume that the evanescent light coming from the SiN waveguide can be split into three distinct paths:

1. Absorption in the Si volume, contributing to detectable avalanches.
2. Loss to the surrounding medium (radiative scattering or parasitic absorption).
3. Transmission through the SiN waveguide, representing unabsorbed signal.

Similarly, the Si SPAD volume is partitioned into three regions (Fig. 1): the central volume or core region (green), the sensitive region (red) which includes the central region and the side volume or ribs, and the lossy volume or contact region (grey). Absorption is intended to be

maximized in the core volume where the SCR will be defined. Absorption in the ribs is still considered to contribute to the avalanche events via carrier diffusion, albeit with a reduced efficiency. The absorption taking place in the contact region is considered to be a loss and does not contribute to avalanche events. This is why the sensitive region is defined as the union of the central and side volumes. Radiative losses are monitored at the simulation domain boundaries, while transmitted power is measured at the SiN waveguide output (purple). Simulations are performed assuming a wavelength $\lambda = 600$ nm, targeting typical diamond color center emission wavelengths [20]. For our optical simulation, the diode length is varied from 20 to 200 μm , with a fixed 50 nm oxide gap between the Si and SiN waveguides. For comparison, a non-etched geometry is also modeled - denoted as the 'planar device'.

FDTD results show how absorption scales with diode length, and highlight the differences between planar and etched device structures (Fig. 2(a),(b)). A graphical visualization of how the absorption is distributed throughout the structures is presented in Fig. 2(c) and (d), where the absorption probability, P_{abs} , is shown integrated over the waveguide z-height (2-D maps, bottom) and over both height and length (1-D line plot, top). As evidenced by the presented Fig. 2, due to diffraction, absorption saturates at a shorter length (between 125 μm and 150 μm) in the planar device compared to the etched one (at 200 μm). Although absorption remains highest near the center, only 50% of the total radiation is confined in the core region. In contrast, the etched structure demonstrates superior performance for this application as it is more efficient concentrating most absorption within the core where up to 90.8% of the radiation is absorbed. This efficient coupling scheme appears well-suited for the target wavelength and requires minimal optimization. Transmission loss is kept acceptably low for both SPAD designs with a sufficiently long diode. The spatial absorption profile, the key output of this analysis, is used for the subsequent calculation of PDE. Given the clear advantage of the etched structure in absorption concentration and efficiency, only this architecture is considered for further steps.

2.2. TCAD simulations

Configuring the single-photon avalanche diode (SPAD) as a rib waveguide effectively localizes the optical absorption volume within the device core. However, the overall SPAD performance derives not only from efficient optical absorption but also from its electrical performance. Dark counts, for one, are completely independent on the device's optics, and similarly is the avalanche efficiency. Therefore, Technology Computer-Aided Design (TCAD) simulations are utilized to estimate the breakdown voltage (V_{BD}), assess metrics such as electric field distribution and calculate the avalanche trigger efficiency across the waveguide's cross-section. While a simplified 1-D simulation—considering only electric fields normal to the junction—would reduce computational cost, a 2-D approach is adopted to account for the significant (transverse to the propagation direction) field components inherent to shallow rib waveguides. The 2-D model assumes uniform diode characteristics along the device length, enabling efficient 3-D performance analysis without excessive computational burden. The doping profile, derived from the foundry-provided process flow is simulated using Sentaurus SProcess software, and is illustrated in Fig. 3(a) and (c), with the electric field cutline shown in d). Due to high implantation energies used, the resulting doping profile, with associated straggle, is centered closer to the bottom interface of the Si waveguide. Subsequent fast annealing steps further accentuate this effect, as dopants from higher concentration regions diffuse towards the core through the rib sections, increasing the doping contrast found between the bottom interface and the top. Carrier transport in the device is described by the drift-diffusion equation. Fermi-Dirac statistics are accounted for in defining the electron and hole distribution in the silicon volume. After careful process simulation, an electrostatic analysis on the SPAD is performed to examine its feasibility to operate in a Geiger mode device. One of the results is the V_{bd} , which is calculated as to be 13.3 V (300 K). Surface trap effects are modeled following methodologies developed for

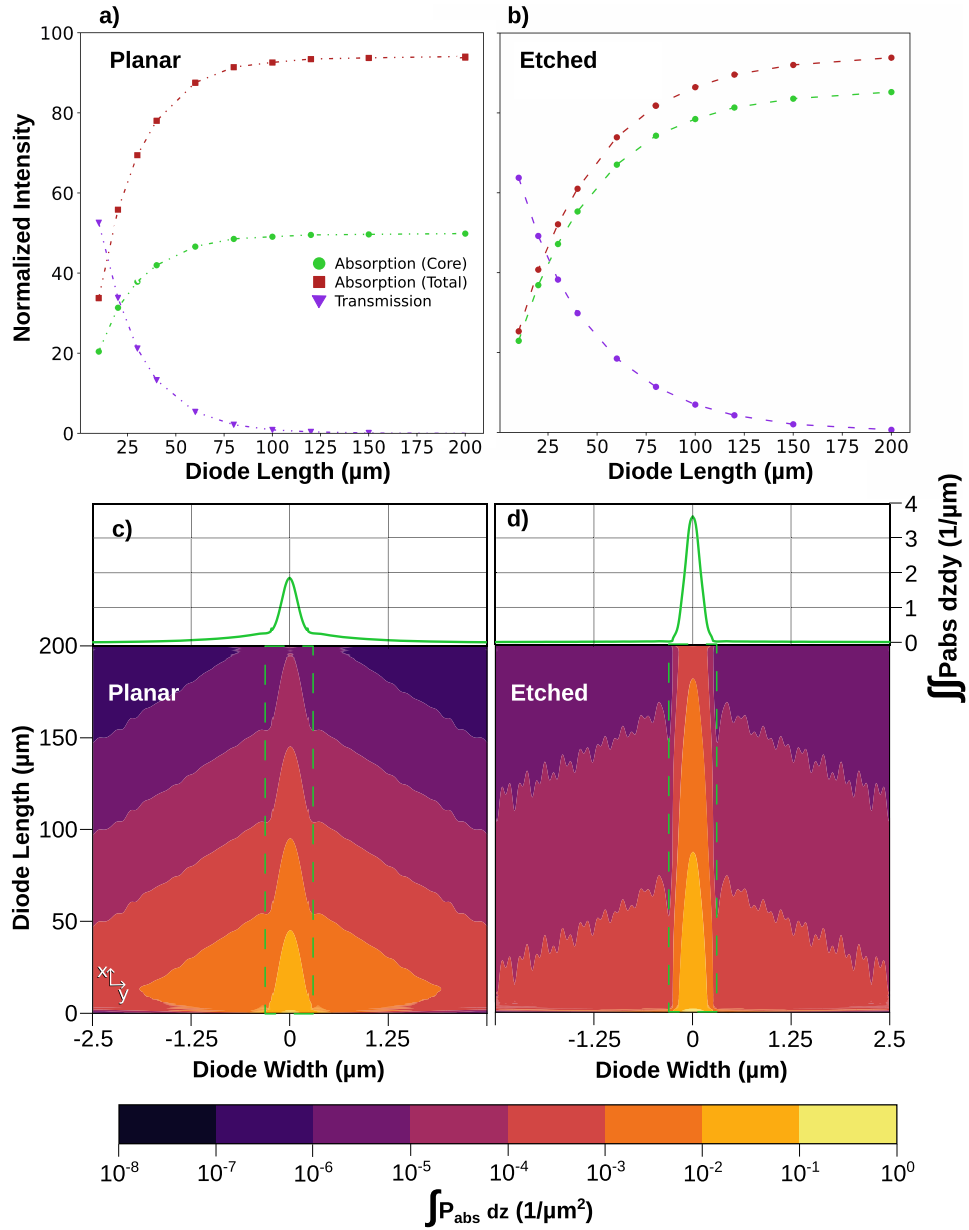


Fig. 2. FDTD model output for etched and planar Si-SPAD. Length dependent absorption plots for a) planar and b), illustrate how the etched waveguide allows for a higher confinement of the absorption volume, with more than 80% of the radiation being absorbed in the core area, compared to just over 50% for the planar case (when considering a 200 μm device). c) and d) show the absorption probability integrated over one- and two-dimensions. In the bottom contour plots, P_{abs} per unit area is shown, integrated over the waveguide height. Above, integrated over height and length, the P_{abs} per unit width, in green, is shown. This stresses how the etched diode exhibits stronger optical confinement compared to planar device, and how it is better suited for SPAD applications.

imec's isipp50g platform [21]. In this work, trap concentrations were deduced from carrier lifetime measurements using transient two-photon absorption techniques, ensuring accurate representation of surface related trap states. These are modeled in our TCAD simulation following the aforementioned work, where the electron and hole cross sections are defined as 2.1×10^{-15} and $2.6 \times 10^{-16} \text{ cm}^{-2}$, respectively, and the trap energy level is assumed to be near the middle of the bandgap, 0.5 eV above the valence band. The ionization coefficients for each carrier type (α_e , α_h) and the electron-hole pair (EHP) generation rates for the Shockley-Read-Hall and band-to-band tunneling mechanisms (G_{SRH} , G_{B2B}) are calculated in this step. These calculations are performed across a range of voltages, extending from below to above the breakdown voltage (V_{bd}). The excess bias V_{ex} is expressed as a percentage relative to V_{bd} . The method begins with the description of the carrier transport in the structure, by solving the drift-diffusion equation, accounting for Fermi-Dirac statistics. Doping-dependent carrier mobility is modeled using the Masetti model, which incorporates the most current empirical data [22]. The aforementioned $\alpha_{e,h}$ coefficients that describe the charge multiplication were calculated according to the New University of Bologna Model [23–25]. An impact ionization model that considers both local and non-local electric fields and thus accounts for the dead-space effect was chosen. This is important given the relatively thin multiplication region considered here, where the dead space size will cover a significant portion of the region's width [26,27]. The generation-recombination rates in the Shockley-Read-Hall model are influenced by deep defect levels in the bandgap, with carrier lifetimes τ_e and τ_h modeled as functions of doping using the Scharfetter model [28]. Trap assisted tunneling (TAT) effects are included in the simulation by the non local path Schenk model, to better fit to the rather abrupt, high field junction modeled here [29]. The phonon assisted band-to-band tunneling effect is modeled with the Schenk model, which takes into account field-enhancement, and translates the carrier generation enhancement due to the thinning of the potential barrier along the multiplication region when the diode is operated in Geiger mode [30]. The default parameter values provided by the TCAD tool were unaltered for the mentioned physical models [31].

This TCAD simulation provides the key electrical parameters required to estimate the quality of a SPAD such as the spontaneous generation rates of carriers and the spatial distributions of the field and ionisation coefficients α_e and α_h for the electrons and holes respectively.

2.3. Post-processing

From the TCAD simulation, the avalanche triggering probability can be estimated. This is relevant as, to result in a detection, the generated electron-hole pair (EHP) must trigger an avalanche, as described by analytical models commonly used when designing SPADs for CMOS technologies [32]. Together with the absorption probability map, the PDE can be deduced. Those carrier specific triggering probabilities (P_e , P_h) are analytically calculated from the α coefficients by solving the following differential equations, integrating over the electrical field lines [33]:

$$\begin{cases} \frac{dP_e}{ds} = (1 - P_e)\alpha_e(P_e + P_h - P_e P_h) \\ \frac{dP_h}{ds} = -(1 - P_h)\alpha_h(P_e + P_h - P_e P_h) \end{cases} \quad \text{with boundary condition} \quad \begin{cases} P_e(s = s_n) = 0 \\ P_h(s = s_p) = 0 \end{cases} \quad (1)$$

where s , denotes the (x, y, z) coordinates within the space charge region of the diode. For this simulation approach, the TCAD simulation is performed for the device in the y and z axis, as along the x axis (diode length), uniform conditions are assumed. Moreover, α_e , α_h are the ionization coefficients resulting from the device simulation. The coordinates s_p , s_n constitute the boundaries of the SCR at the p and n sides respectively. The avalanche triggering probability (P_{tr}) at position s can be numerically obtained as proposed in [34]. It conveniently results from a

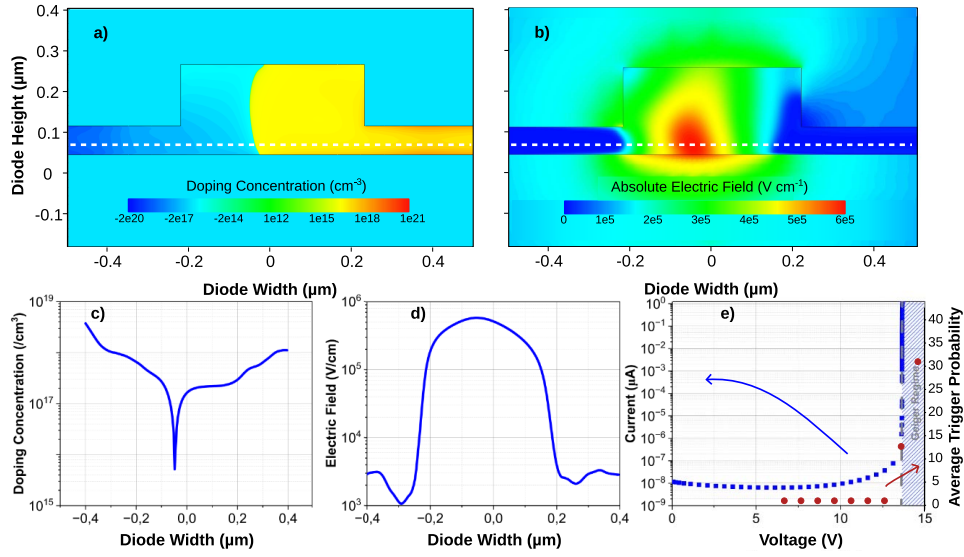


Fig. 3. a) 2D contour map of the doping profile of the etched SPAD extracted from TCAD process simulation using the foundry implantation conditions. b) 2D contour map of the electric field distribution at the breakdown voltage obtained from TCAD device simulation. Line profiles of c) doping concentration and d) electric field along the cutlines indicated in a) and b), respectively. The boundary conditions, s_p and s_n , referred in (1) are denoted in the figure e) Simulated reverse I–V characteristics and ATP, highlighting the Geiger-mode operating region of the SPAD.

contribution of the triggering probabilities by electrons P_e and holes P_h so that:

$$P_{tr}(s) = P_e + P_h - P_e P_h \quad (2)$$

The average trigger probability (ATP) is also calculated and serves as a way to confirm V_{bd} , as such:

$$ATP = \frac{1}{s_n - s_p} \int_{s_p}^{s_n} P_{tr}(s) ds \quad (3)$$

Both the current rise and the first non-zero ATP value coincide in the same voltage region, as shown in 3(e) ensuring a consistent evaluation of V_{bd} . The photon detection efficiency (PDE) is defined as the probability that a photon of a specific wavelength (i) couples to the waveguide SPAD, (ii) is absorbed in the active region, and (iii) triggers a detectable macroscopic avalanche. To model this process, we decompose the PDE into sequential steps. First, the photon must be absorbed in the sensitive region—either the space-charge region (SCR) or adjacent quasi-neutral regions. This absorption probability, modeled using finite-difference time-domain (FDTD) simulations (see Subsection 2.1), provides a spatial map $P_{abs}(s)$ (given in cm^{-3}). This calculation adapts established analytical frameworks [32] to the waveguide geometry.

The multiplication region is centered in the waveguide core and comprises four distinct zones: (1) the SCR in the core, (2) neutral p/n regions in the core, (3) neutral regions in the rib, and (4) highly doped silicon/silicide contacts. Carriers generated in the contact regions are lost to recombination, as high doping and defect concentrations prevent collection. As such, absorptions in the contacts are excluded, while those in the active region fall into two cases:

Case (1): Absorption in the SCR. The EHP is accelerated by the high electric field, directly triggering an avalanche. The PDE contribution is:

$$PDE_{(1)} = \int_{s_p}^{s_n} P_{tr}(s) P_{abs}(s) ds \quad (4)$$

where $P_{tr}(s)$ is the avalanche triggering probability. As the TCAD simulation is carried out in 2-D, $P_{tr}(s)$ is constant over the length of the diode.

Case (2): Absorption in neutral regions. Minority carriers diffuse to the SCR edges, where they trigger avalanches with probability $P_{tr}(s_{n,p})$. Bulk recombination losses are negligible due to the picosecond-scale diffusion time versus nanosecond-scale carrier lifetimes [21,35]. Surface recombination at the Si/SiO₂ interface, governed by:

$$\phi_{h,e}|_{boundary} = (p, n - p_0, n_0) R_{surf}, \quad (5)$$

is also negligible, as even extreme surface recombination velocities (R_{surf}) do not impede carrier collection. Thus, the collection efficiency $\eta_{e,h}(s)$ can be considered as unity, giving the second and third contribution to PDE as:

$$PDE_{(2)} = P_{tr}(s_n) \int_{s_{nc}}^{s_n} P_{abs}(s) ds \quad (6)$$

and

$$PDE_{(3)} = P_{tr}(s_p) \int_{s_p}^{s_{pc}} P_{abs}(s) ds, \quad (7)$$

with the coordinates s_{pc} and s_{nc} the boundaries of the p- and n-side contacts, respectively. The total PDE combines all contributions:

$$PDE = PDE_{(1)} + PDE_{(2)} + PDE_{(3)} \quad (8)$$

The PDE and dark count rate (DCR) both depend on the excess bias V_{ex} . The DCR arises from thermally or field-generated EHPs in the SCR, modeled as:

$$DCR = \int_{s_p}^{s_n} G_{EHP}(s) P_{tr}(s) ds, \quad (9)$$

where $G_{EHP}(s)$ includes Shockley-Read-Hall (G_{SRH}) and band-to-band (G_{B2B}) generation rates from TCAD simulations. For the calculation done in Eqs. (4), (6) and (7), this is performed numerically by first computing the absorption probability density (P_{abs}), with units of μm^{-3} and the avalanche triggering probability (P_{tr}), for each slice in the yz-plane. Each 2D slice is then multiplied point-by-point, and the resulting values are numerically integrated over the length of the diode (the x-axis) to reconstruct the total PDE for the volumetric device. This ensures that the overall calculation correctly accounts for the per-unit-volume nature of the absorbed power, and that the result is dimensionless, as required. An example of the Shockley-Read-Hall recombination-generation and band-to-band generation map is shown in Fig. 3 at an excess bias of 5%. Here, for G_{SRH} , the effect of surface trap states can be seen to contribute to generation in the Si/SiO₂ boundaries. Due to the geometry of the proposed detector, the DCR is expressed in Hz/ μm (relative to the diode length) rather than Hz/ μm^2 , as is typical for traditional SPAD geometries where the pixel area is the relevant parameter.

3. Results and discussion

In this section, the simulation results pertaining the proposed architectures are presented and discussed. First an analysis of the electrostatic behaviour of the chosen doping schemes is carried out. This is followed by the discussion of the main figures of merit of the SPAD: the P_{tr} , the DCR and the PDE. These results are experimentally verified by analysing the measured DCR of existing devices and comparing it to the simulated DCR.

3.1. Electrostatic analysis, breakdown voltage and avalanche triggering probability

An electrostatic analysis on the waveguide SPAD has been conducted in order to validate the avalanche diode architecture for a correct Geiger-mode operation. Figure 3(a) and (c) show respectively the 2-D doping profile and the corresponding 1-D line profile along the white dashed line, obtained from the process simulation. The profiles indicate a smooth doping distribution in the SPAD, as desirable for good device operation. In Fig. 3(b) and (d), the 2-D profile and a 1-D cross-section of the electrical field are shown for a bias just above the breakdown voltage (5% overbias). Here, the avalanche generation model is switched off, to model the pre-avalanche metastable regime, right before the multiplication process takes place. This allows to observe the electric field without being disturbed by the high amount of charges that would otherwise flow through the junction during avalanche. In the 1-D plot of Fig. 3(d), a uniform field is observed in the active region, which is key for a uniform avalanche trigger probability. The field sharply drops towards the rib regions, thanks to the smooth doping profile, ensuring that premature edge breakdown (PEB) does not occur, and ensuring compatibility with Geiger-mode operation. While PEB is prevented, a small non-uniformity in the field persists along the diode height, correlating with the also non-uniform doping profile described earlier in section 2.2. To define the operation point where Geiger-mode operation is possible for the considered SPADs, the bias voltage was swept in the TCAD software. The resulting curve can be found in Fig. 3(e). The magnitude of the current values has no physical significance, as an adaptive external resistor method was put in place to improve the convergence of the model.

3.2. Noise performance and experimental verification

The noise of the simulated device is mainly attributed to two spontaneous EHP generation mechanisms, i.e., G_{SRH} and G_{B2B} . The 2-D maps of the G_{SRH} and G_{B2B} contributions at 5% excess bias are shown in Fig. 4(a) and (c). Figure 4(b) presents the simulated DCR due to G_{SRH} and G_{B2B} with respect to applied excess bias. It is evident that the noise of the device is mainly governed by G_{SRH} , while the contribution from G_{B2B} is negligible ($< 1 \text{ Hz}/\mu\text{m}$) at these biases, and is concentrated on the high field zone near the bottom of the waveguide. However, both mechanisms' contributions scale with field strength, and G_{B2B} is set to increase in relative significance for higher V_{ex} .

To validate our simulation model, the DCR of a waveguide PN diode, corresponding to the etched geometry depicted in Fig. 1 has been measured. This device was fabricated in imec's isipp50G. platform. To do so, a gated Geiger mode setup was devised, where the device under test (DUT) is assembled on a biasing PCB, as shown in Fig. 4(e). The DUT is placed in a dark chamber with thermoelectric cooling for temperature control. A periodical gated signal is applied to the DUT, and the response is monitored by an event counter (IDQuantique's ID1000 Time Controller), which is synchronized to the gate signal frequency. A low frequency of 50 kHz is used to avoid afterpulsing from affecting the measurement results. The diode is operated in a gated mode, quenching the avalanches, and gate time is large enough (30 ns) so that effective V_{bd} matches DC conditions [36], measured to be 11.3 V. This measured value is 2 V smaller than the simulated one, a gap of approximately 20 %. This is recognized as a non-trivial discrepancy, that is attributed to a few points. There exists possible sources of premature breakdown such as the presence of edged facets covered with SiO_2 at both ends of the diode. There can also be discrepancies due to fabrications (few percents) uncertainties on width of the silicon, position accuracy of the implantation or even sheet resistance of the silicon layer. Nonetheless, the presented analysis captures the essential physics and provides a good agreement given the information the authors have access to. The measured dark events are translated into the equivalent free running DCR following the methodology previously described in [11]. The measured DCR of the device was compared with simulation, obtained by replicating the process flow described by the foundry and following the steps described in 2 section. This is shown

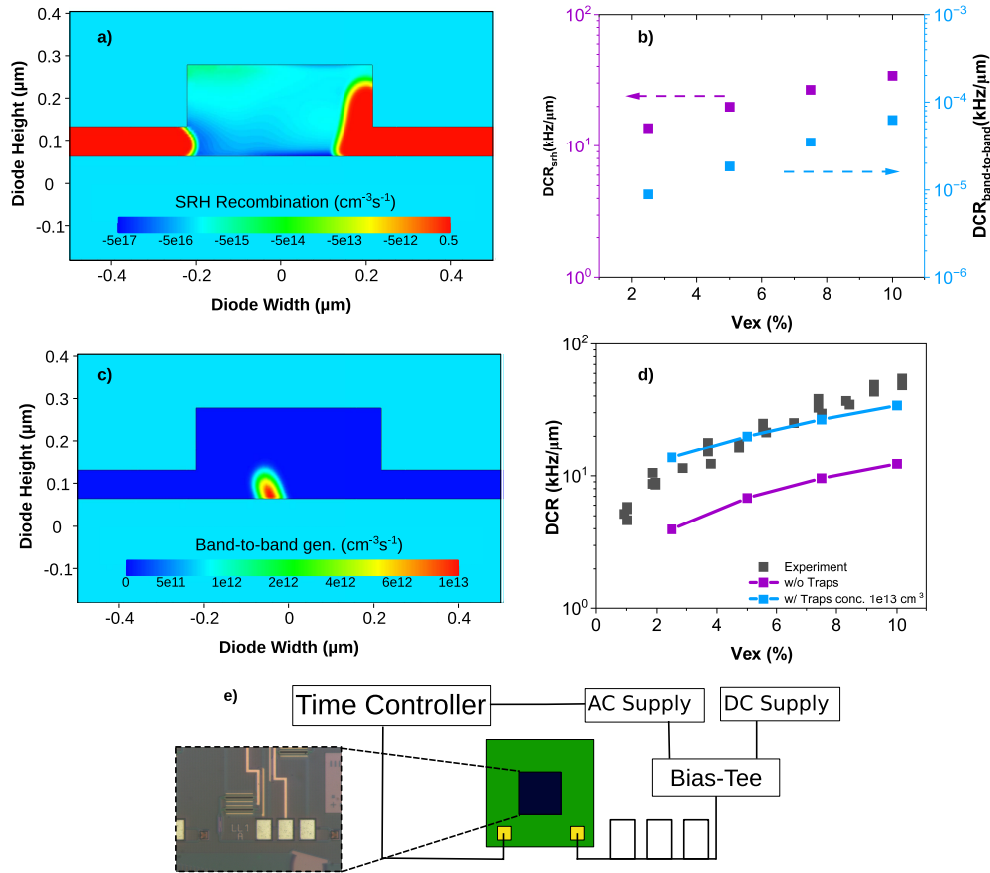


Fig. 4. Dark counts in the fabricated and the simulated Si-SPAD; a) Shockley-Read-Hall recombination rate map showing the charge carrier generation is observed throughout the depletion region, with a marked contribution on the Si/SiO₂ interfaces due to surface traps; b) Band-to-band tunneling generation, which is shown to contribute to the overall DCR but at much lower rates than SRH, as corroborated by c). d) experimental values of the DCR values and theoretical values obtained through simulation. e) Schematic of measurement setup.

in Fig. 4(d). Good matching between simulation and experiment is achieved when including surface trap effects. A surface trap concentration of 10^{13} cm^{-2} is considered, aligning with those previously deduced from two-photon absorption measurements [21], as previously discussed. This validates the proposed model as an effective tool to predict device performance.

3.3. PDE performance and overview

To comprehensively evaluate the device performance in terms of detection efficiency, a device with $200 \mu\text{m}$ length was further simulated under excess biases up to 50 %, and the results are presented in Fig. 5. The photon detection efficiency (PDE) of a SPAD is perhaps its most relevant metric, and depending on V_{ex} and diode length, PDEs in this configuration can reach 76% (Fig. 5(a)). For high V_{ex} values such as 50 %, it can be seen (5(b)) that band-to-band tunneling becomes more prevalent, reaching $10^3 \text{ kHz}/\mu\text{m}$. Even though it remains a much lower contribution relative to SRH, it remains important to keep in mind for future optimization efforts. In future devices where SRH is suppressed and surface trap contributions minimized,

band-to-band tunneling can then become relatively more impactful. We see an increase of 5 orders of magnitude from the 10 % V_{ex} value (Fig. 5(b)) to the 50%. The high DCR at high excess biases emphasizes the need for doping architecture optimization. Similarly to top illuminated CMOS fabricated Si-SPADs, junction engineering and improving material quality can lead to marked improvements on the device performance metrics, by virtue of factors such as reducing trap density, working to isolate and further uniformize the high-field region and also on the widening of the active region [37,38]. The current device faces a fundamental challenge in operating above 10 % V_{ex} . At higher excess biases such as 20%, the dark count probability escalates dramatically due to enhanced band-to-band tunneling and trap-assisted generation mechanisms. To maintain dark count probabilities below acceptable thresholds (e.g., <10 %), the gate window would need to be reduced to unreasonable gating times, as per [11]. This limitation is critical, as it locks the > 50 % PDEs that are, theoretically, achievable with this architecture behind the prohibitive DCR figures. This fundamentally underscores the need to optimize the device architecture specifically for reduced DCR.

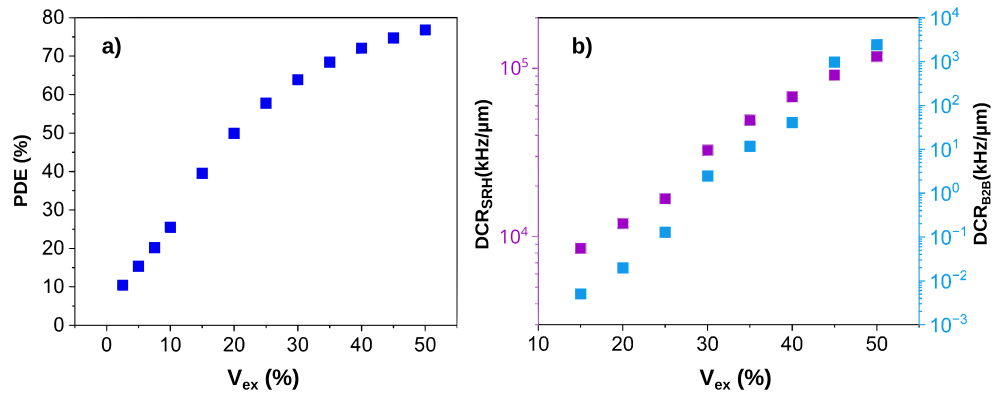


Fig. 5. Simulated a) photon detection efficiency and b) dark count rate for a 200 um long diode. PDE tapers out at 76%. Band to band tunneling increases in relevance for higher excess biases.

Finally, while the presented model does not address the topic of jitter and afterpulsing, it should be expected for waveguide integrated SPADs for this to be quite an advantage, as posited by previous works [7]. Previous measurements of vertically illuminated planar geometry SPADs saw an upper limit of 160 ps, limited by test equipment [39]. The added optical mode confinement and resulting overlap with SCR on evanescently coupled waveguide SPADs should only result in smaller jitter values. On the subject of afterpulsing, given the decay dynamics of silicon waveguides presenting fast, sub ns and long, >100 ns stages [21], such waveguide integrated SPADs are expected to exhibit long afterpulsing tails. This should align with other observations in literature, where long trap lifetimes are correlated with measurable afterpulse tails, and that optimization of silicon quality and operating with appropriate device hold off times or frequency (for free running or gated modes of operation) will be crucial for future work [40–42].

4. Conclusion and future prospects

The findings reported here reveal the potential of waveguide-SPADs for integrated photonics using an off the shelf, readily available integrated photonics platform (isipp50G). By reconfiguring a conventional PN waveguide modulator for Geiger-mode operation and leveraging micro-transfer printing techniques, we propose a pathway to integrate high-efficiency SPADs with SiN photonic platforms for visible-wavelength quantum applications. Extensive work remains ahead, for which the presented methodology can serve for design technology co-optimization (DTCO)

of hybrid integrated SPADs with visible photonics-enabling platforms. We highlight the most critical future steps to be on developing optimized doping architectures (vs taking a pre-existent architecture optimized for injection based modulation), with a main concern on minimizing DCR. Guaranteeing a junction design where the active region is defined over a P⁻/N combination will enable higher efficiencies at lower excess biases by employing the high multiplication efficiency of electrons, compared to the current design where both P and N sides are of similar doping concentration. Moreover, with current doping levels, band-to-band generation is non-negligible. Thus, by lowering the overall p-type doping concentration, an improved DCR should be achievable by minimizing the band bending at the junction from the large dopant concentration contrast. Furthermore, it was shown that trap-assisted generation contributes significantly to DCR, and as such strategies to confine the active region away from the Si/SiO₂ regions should be implemented. Addressing the observed DCR limitations through design-technology co-optimization, the superior PDE performance predicted by the model can be unlocked, allowing reaching its full potential for integrated quantum photonic applications. Despite the currently high DCR value at room temperature, the future seems bright for single-photon detection in the field of visible integrated photonics, with waveguide-integrated Si-SPADs presenting themselves as a viable CMOS-compatible and temperature robust alternative.

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